

This exam has 6 pages and 8 exercises.

The result will be computed as (total number of points plus 10) divided by 10.

1. Island of liars and truth speakers (11 points)

Suppose you ask the following question to an islander hurrying by on a T-crossing to find the way to the harbour:

Is the declarative sentence “*you are a truth speaker $\leftrightarrow$ the harbour is to the right*” true?

Either explain that this question suffices to determine the way to the harbour, or explain why this is not the case.

Solution: This question indeed suffices to determine the way to the harbour. There are four cases to distinguish: the islander is a truth speaker or a liar, and the harbour is to the right or to the left.

Suppose the islander is a truth speaker. If the harbour is to the right, the islander will say yes, if the harbour is to the left, the islander will say no.

Suppose the islander is a liar. If the harbour is to the right, the declarative sentence is false, so the islander will say yes. If the harbour is to the left, the declarative sentence is true, so the islander will say no.

Concluding, if the islander says yes, the harbour is to the right, and if the islander says no, the harbour is to the left.

2. DPLL procedure (11 points)

Apply the DPLL procedure to the CNF

$$(p \vee \neg q \vee \neg r) \wedge (\neg p \vee q \vee \neg r) \wedge (\neg p \vee \neg q \vee \neg r) \wedge r$$

to check whether it is satisfiable. Explain in detail how the procedure comes to a result. Let the procedure branch first on p , then on q , and then on r ; and let it always try the truth value true ($\top$) before false ($\perp$).

Solution: First, substitute $\top$ for r , because of the unit clause r . The three occurrences of $\neg r$ in the three first disjunctions then reduce to $\top$ and can be eliminated, yielding

$$(p \vee \neg q) \wedge (\neg p \vee q) \wedge (\neg p \vee \neg q)$$

Substituting $\top$ for p , reducing the first disjunction to $\top$, and reducing the occurrences of $\neg p$ in the other two disjunctions to $\perp$, yields

$$q \wedge \neg q$$

Substituting $\top$ for q yields $\top \wedge \neg \top$, which reduces to $\perp$. Now we backtrack and substitute $\perp$ for q , yielding $\perp \wedge \neg \perp$, which again reduces to $\perp$.

Now we backtrack and substitute $\perp$ for p . Reducing the last two disjunctions to $\top$, and reducing the occurrence of p in the first disjunction to $\perp$, yields

$$\neg q$$

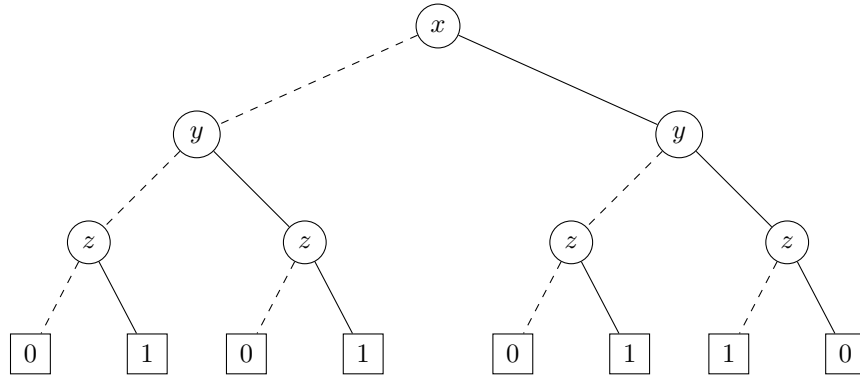
Since this is a unit clause (or because q only occurs negatively), we substitute $\perp$ for q , yielding $\neg \perp$, which finally reduces to $\top$.

So we find that substituting $\perp$ for p and q and substituting $\top$ for r yields a satisfaction of the original formula.

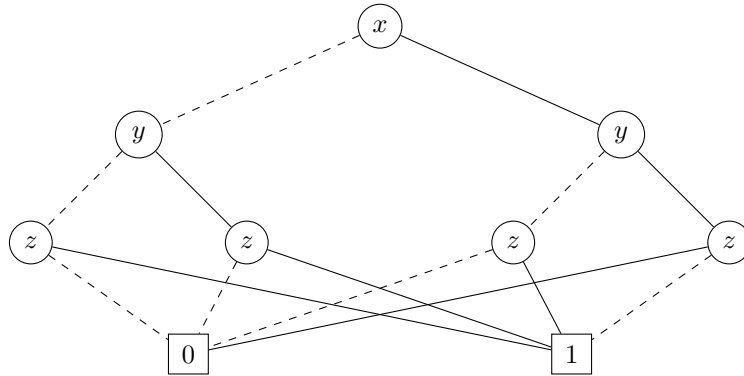
3. OBDD (3+9 points)

- Represent $(x \wedge y) \oplus z$ by means of a binary decision tree, with respect to the variable ordering x, y, z .
- Reduce this binary decision tree to an ordered binary decision diagram. (Also give the intermediate reduction steps.)

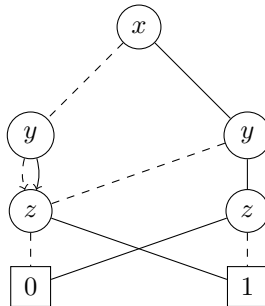
Solution:



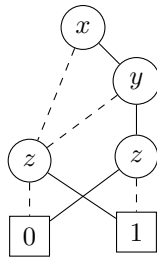
Apply C1 to collapse leaves.



Apply C3 to collapse the three left-hand nodes z :



Apply C2 to remove the left-hand node y :



4. Predicate logic (11 points)

Give two models to show that the formulas $\forall x (C(x) \rightarrow D(x))$ and $\forall x (D(x) \rightarrow C(x))$ do not semantically entail each other. (Explain your answer.)

Solution: Let model $\mathcal{M}_1$ consists of a single element a , where C holds for a while D does not. $\forall x (C(x) \rightarrow D(x))$ does not hold on this model because the universal quantification fails for a , while $\forall x (D(x) \rightarrow C(x))$ holds on this model because D does not hold for the single element a . This shows the second formula does not semantically entail the first one.

Vice versa, let model $\mathcal{M}_2$ consists of a single element b , where C does not hold for b while D does hold for b . $\forall x (C(x) \rightarrow D(x))$ holds on this model because C does not hold for the single element b , while $\forall x (D(x) \rightarrow C(x))$ does not hold on this model because the universal quantification fails for b . This shows the first formula does not semantically entail the second one.

5. Sets (6 points)

Assuming that $\#U = 16$, $\#A = 6$, $\#B = 8$, $\#C = 7$, $\#(A \cap C) = 3$, $\#(A \cup B' \cup C) = 13$, compute the number of elements in $(A \cup C) \setminus B$.

Solution:

$$\begin{aligned} \#((A \cup C) \setminus B) &= \#(A \cup C) \cap B' \\ &= \#(A \cup C) + \#B' - \#(A \cup C \cup B') \\ &= \#A + \#C - \#(A \cap C) + \#U - \#B - \#(A \cup C \cup B') \\ &= 6 + 7 - 3 + 16 - 8 - 13 = 5. \end{aligned}$$

6. Ordering+Equivalence Relations (3+6+4+5 points)

In the set $V = \{a, b, c, d, e\}$ consider the relation R with matrix representation

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \end{bmatrix}$$

- List all elements of the relation R .
- Show that R is an equivalence relation.
- Prove that R is not an ordering relation and not a function.
- Write down the equivalence classes and a system of representatives for R .

Solution:

- a) $R = \{\langle a, a \rangle, \langle a, e \rangle, \langle b, b \rangle, \langle b, c \rangle, \langle c, b \rangle, \langle c, c \rangle, \langle d, d \rangle, \langle e, a \rangle, \langle e, e \rangle\}$.
- b) R is reflexive, because there are only 1's on the diagonal. It is symmetric because the matrix is symmetric around the diagonal. It is transitive because the relations between a and e and between b and c are the only non-reflexive relations.
- c) R is not an ordering relation, since it is not anti-symmetric (e.g. $\langle a, e \rangle, \langle e, a \rangle \in R$).
 R is not a function, since there is an element (e.g. $a \in R$) that relates to more than one element ($\langle a, a \rangle, \langle a, e \rangle \in R$).
- d) The equivalence classes are $\{a, e\}, \{b, c\}, \{d\}$ and a system of representatives for R is e.g. $\{a, b, d\}$.

7. Functions (6+4 points)

We are given the functions $f : \mathbb{R} \rightarrow \mathbb{R}$, $g : \mathbb{R} \rightarrow \mathbb{R}$, $mul_2 : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \sqrt{|x| - 1}, \quad g(x) = \frac{1}{x^2 + 1}, \quad \text{and} \quad mul_2(x) = 2x.$$

- a) For each function, determine its domain of definition and its range.
- b) Give a description (and expression for $h(x)$) for $h = g \circ mul_2 \circ f$ and check whether h is injective.

Solution:

- a) $D_f = (-\infty, -1] \cup [1, +\infty)$, $R_f = [0, +\infty)$, $D_g = \mathbb{R}$, $R_g = (0, 1]$, and the domain of definition and range of mul_2 is $\mathbb{R}$.
- b) $h(x) = (g \circ mul_2)(\sqrt{|x| - 1}) = g(2\sqrt{|x| - 1}) = \frac{1}{4(|x| - 1) + 1}$ and h is not injective, since e.g. $h(-1) = \frac{1}{4} = h(+1)$.

8. Induction and recursion (3+8 points)

We claim that the following statement is true for all natural numbers $n \geq 1$:

$$\sum_{k=1}^n 2(k-2) = n^2 - 3n.$$

- a) Verify by explicit computation that the claim is true for $n = 1$, $n = 2$ and $n = 3$.
- b) Prove by mathematical induction that the statement holds true for all natural numbers n .

Solution:

$$\text{a) } \sum_{k=1}^1 2(k-2) = -2 = 1^2 - 3, \sum_{k=1}^2 2(k-2) = -2 = 2^2 - 3 \cdot 2, \sum_{k=1}^3 2(k-2) = 0 = 3^2 - 3 \cdot 3.$$

$$\text{b) Base case: } \sum_{k=1}^1 2(k-2) = -2 = 1^2 - 3.$$

Induction step: Let $m \geq 1$. Assume that $\sum_{k=1}^m 2(k-2) = m^2 - 3m$ (IH). Then we have

$$\begin{aligned} \sum_{k=1}^{m+1} 2(k-2) &= \sum_{k=1}^m 2(k-2) + 2((m+1)-2) \\ &\stackrel{\text{(IH)}}{=} m^2 - 3m + 2((m+1)-2) \\ &= m^2 - m - 2, \\ (m+1)^2 - 3(m+1) &= m^2 + 2m + 1 - 3m - 3 \\ &= m^2 - m - 2. \end{aligned}$$

Hence (IH) implies $\sum_{k=1}^{m+1} 2(k-2) = (m+1)^2 - 3(m+1)$.